AI REASONING ASSIGNMENT

A Unified Classical Reasoning Agent

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Project summary

This project implements a single prediction interface for three kinds of reasoning problems. Natural-language knowledge bases and queries are converted into symbolic representations. The final decisions are produced by classical reasoning procedures rather than by a generative model. The system uses Horn-clause reasoning for RuleTaker, first-order resolution for FOLIO, and structural causal model reasoning for CounterBench.

3	77,230	45
public datasets	resolved public predictions	unresolved examples logged separately

Prepared from the executed public-evaluation notebook and its exported public prediction files. The measurements below are the recorded run outputs, not a new evaluation performed while preparing this report.

1. Problem and system design

The task is to decide whether a query follows from a knowledge base. The system follows the required pipeline: natural-language input -> internal representation -> classical inference -> Boolean prediction. It exposes one common prediction entry point and selects a reasoning module using input structure and content cues; an explicit dataset name is not required by the prediction call.

1.1 Dataset coverage

Dataset	Public split / files used	Examples	Internal representation
RuleTaker	tasksource/ruletaker - dev	75,872	Function-free predicates, facts and Horn rules
FOLIO	tasksource/folio - validation	203	First-order formula trees and clauses
CounterBench	Alpha V1 and Backdoor V2 JSON files	1,200	Structural causal equations, observations and interventions

1.2 Unified prediction flow

Each example is inspected for recognizable fields and logical/causal structure. The router selects a symbolic reasoning path. The module returns a prediction and diagnostic status. During public evaluation, gold labels and source/dataset metadata are removed from the example passed to the predictor; labels are retained separately only for metric calculation.

Input knowledge base + query	Routing schema and content cues	Reasoning module Horn / FOL / causal	Output True or False

2. Symbolic representations and algorithms

2.1 RuleTaker: Horn-clause reasoning

Facts and rules are represented as predicate symbols with argument tuples. Variables are marked explicitly, and rule bodies may contain positive and negative literals. The main public evaluation uses semi-naive forward chaining: only newly derived facts are used to trigger the next inference round. Predicate/argument indexes reduce candidate matching work. Unification binds variables to constants; stratified negation-as-failure checks whether a positive literal can be proved before allowing its negation. Backward chaining is also implemented as an alternative engine and is exercised by the self-tests.

2.2 FOLIO: first-order logic

FOLIO examples are represented as first-order formula trees. The implementation separates concatenated premises, parses quantifiers and connectives, performs Skolemization and conversion to conjunctive normal form, then applies unification, factoring and binary resolution. The prover checks both the conclusion and its negation. FOLIO has a three-valued source label (True, False, Uncertain); for the assignment’s binary output, Uncertain is mapped to False when a prediction is available. A proof result that remains unresolved is kept as unknown rather than fabricated.

2.3 CounterBench: causal reasoning

Causal statements are translated into structural causal mechanisms. The module records signed causal parents, endogenous variables, exogenous/root variables and available Bernoulli probabilities. Observations constrain the admissible exogenous assignments (abduction). The intervention is applied to the relevant variable before forward simulation of downstream mechanisms. The predicted outcome is derived from the resulting counterfactual world. Unsupported structures or cases with no compatible assignment remain unresolved.

2.4 Compliance with the classical reasoning requirement

No generative AI model is used to produce True/False decisions. Natural-language conversion is performed by hand-written parsing and normalization code. FOL formulas are converted into symbolic structures using a logic parser. The final decision is made by forward/backward chaining, first-order resolution, or structural causal model simulation. The program does not call a language-model API for label prediction.

3. Public experiments and results

The figures below are copied from the executed notebook run. Accuracy on completed examples is reported separately from coverage so that unresolved examples do not disappear from the evaluation.

Dataset	Total	Completed	Coverage	Accuracy on completed	Coverage-adjusted correct rate	Runtime
RuleTaker	75,872	75,872	100.00%	100.00%	100.00%	~46.3 s
FOLIO	203	168	82.76%	91.67% binary; 83.33% 3-class	75.86% binary	~46.4 s
CounterBench	1,200	1,190	99.17%	94.96%	94.17%	~0.16 s

Interpretation

RuleTaker was evaluated on all 75,872 development examples and all were classified correctly in the recorded run. FOLIO completed 168 of 203 validation examples; among completed cases the recorded three-class accuracy was 83.33%, and binary accuracy after mapping Uncertain to False was 91.67%. CounterBench completed 1,190 of 1,200 cases and achieved 94.96% accuracy on completed cases. Coverage-adjusted rates treat unresolved examples as not correct; they are not the same as accuracy conditional on completion.

The average query time recorded by the notebook was approximately 0.610 ms for RuleTaker, 252.45 ms for FOLIO and 0.133 ms for CounterBench. Runtime depends on hardware, software versions and caching; these are measurements from the recorded execution, not hardware-independent guarantees.

4. Public prediction file

The notebook exported a public-dataset prediction file named submission.csv with the required columns id,label. IDs are dataset-prefixed and use the row position to avoid collisions between source datasets that reuse numeric IDs. The file was checked for duplicate IDs, missing values and invalid labels. The ID format is dataset:row-position so the three public sources can be combined without collisions; preserve the original_id values in the unresolved log for traceability.

Dataset prefix	Rows in CSV	Predicted True	Predicted False	Unresolved
ruletaker	75,872	38,158	37,714	0
folio	168	48	120	35
counterbench	1,190	570	620	10
Total	77,230	38,776	38,454	45

Important coverage note: the CSV contains 77,230 resolved predictions, not all 77,275 public examples. The 45 unresolved examples (35 FOLIO and 10 CounterBench) were written to unresolved_public_examples.csv. They were not assigned arbitrary False labels. If the instructor requires one output for every public example, those unresolved cases must be resolved before treating this CSV as complete.

The private test set was not available or used during development. The notebook retains a separate private-test conversion cell for the evaluator. The public CSV is based on public development/validation examples only. Confirm with the instructor that dataset-prefixed IDs are acceptable for the combined public file; a private evaluator may instead supply its own required ID format.

5. Example reasoning trace

Knowledge base:

```
Bob is young. Bob is kind. If someone is young then they are smart. If someone is smart and kind
then they are successful.
```

Query: Bob is successful.

The Horn engine first unifies young(?x) with young(bob), binding ?x to bob, and derives smart(bob). It then matches both smart(bob) and kind(bob) against the second rule body and derives successful(bob). The query is present in the derived fact set, so the prediction is True.

Derivation:

```
young(bob) [given fact]
smart(bob) [derived from young(?x) -> smart(?x)]
kind(bob) [given fact]
successful(bob) [derived from smart(?x) AND kind(?x) -> successful(?x)]
```

6. Limitations and reproducibility

6.1 Unresolved public examples

The recorded run left 35 FOLIO examples unresolved and 10 CounterBench examples unresolved. The FOLIO diagnostic recorded 12 formal parsing/CNF errors and 23 unresolved proof-search outcomes. The ten CounterBench failures reported that no exogenous assignment was compatible with the supplied observations under the current model assumptions. These cases are listed in unresolved_public_examples.csv.

6.2 Other validation observations

The notebook reports 22/22 self-tests for forward chaining and 22/22 for backward chaining. Eight CounterBench SCM regression tests passed. The FOLIO resolution regression test passed, and automatic-routing checks passed on three representative examples. On 300 generated synthetic Horn problems, forward and backward engines agreed on 298 comparable cases. This agreement check is a diagnostic, not a substitute for benchmark ground truth.

6.3 Reproduction

Run the existing reasoning_agent.ipynb notebook from a fresh kernel, executing cells in order. The notebook downloads the public datasets, runs self-tests, evaluates RuleTaker dev, FOLIO validation and the two CounterBench JSON files, then writes the public submission CSV and an unresolved-case log. Network access is required for the Hugging Face dataset downloads. A Hugging Face token can improve download limits but is not used for inference.

6.4 Dependencies

The notebook uses Python standard-library modules plus pandas, datasets, huggingface_hub and NLTK. The notebook also uses IPython/Jupyter display helpers for tables and progress output. A typical environment can install the external packages with:

```
python -m pip install pandas datasets huggingface_hub nltk ipython jupyter
```

7. Private-test execution

The TA will run the private-test conversion cell after the private evaluation file is supplied. The reasoning agent will generate predictions for the supplied private examples. The private test dataset is reserved for the TA's final evaluation and is not used during development.

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